NARAYANA IIT - JEE ACADEMY - INDIA

Sec: SR.IIT_*COSC(MODEL-A&APEX)	RTA-6	Date: 15-03-26
Time: 3 Hrs	2023_P2	Max. Marks: 180

KEY SHEET

MATHEMATICS

1	**A**	2	**D**	3	**B**	4	**D**	5	**ABC**
6	**ABC**	7	**ACD**	8	**3**	9	**3**	10	**6**
11	**3**	12	**14**	13	**0**	14	**1**	15	**2**
16	**27**	17	**0.50**						

PHYSICS

18	**A**	19	**A**	20	**C**	21	**A**	22	**B**
23	**ACD**	24	**AC**	25	**108**	26	**12**	27	**6**
28	**18**	29	**8**	30	**26**	31	**20**	32	**1**
33	**200**	34	**21**						

CHEMISTRY

35	**B**	36	**D**	37	**A**	38	**B**	39	**BD**
40	**BCD**	41	**ABCD**	42	**6**	43	**4**	44	**5**
45	**6**	46	**4**	47	**18**	48	**7**	49	**7**
50	**0.10**	51	**0.71**						

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SOLUTIONS

MATHS

1. Using the standard binomial summation identity

$$n=\sum_{k=0}^{10}\binom{40+k}{k}\binom{20-k}{10-k}=\binom{61}{10}$$

$$\binom{61}{10}=\frac{61!}{51!10!}$$

The number of trailing zeros equal the Exponent power of 5 in n.

$$E_5(61!)=12+2=14$$

$$E_5(51!)=10+2=12$$

$$E_{10}(10!)=2=2$$

2. Let $S=\sum_{m=1}^{\infty}\sum_{n=1}^{\infty}\sum_{p=1}^{\infty}\frac{1}{\frac{2^m}{m}\frac{2^n}{n}\left(\frac{2^m}{m}+\frac{2^n}{n}+\frac{2^p}{p}\right)}$

Let $\frac{2^m}{m}=a_m,\quad \frac{2^n}{n}=a_n,\quad \frac{2^p}{p}=a_p$

Then $S=\sum_{m=1}^{\infty}\sum_{n=1}^{\infty}\sum_{p=1}^{\infty}\frac{1}{a_m a_n(a_m+a_n+a_p)}$

Interchanging (m, n, p) and adding in cyclic order,

$$3S=\sum_{m=1}^{\infty}\sum_{n=1}^{\infty}\sum_{p=1}^{\infty}\frac{1}{a_m a_n a_p}$$

$$=\left(\sum_{m=1}^{\infty}\frac{1}{a_m}\right)^3=\left(\sum_{m=1}^{\infty}\frac{m}{2^m}\right)^3$$

$$=\left(\frac{1}{2}+\frac{2}{2^2}+\frac{3}{2^3}+\cdots\right)^3=\frac{1}{8}\left(1+2\cdot\frac{1}{2}+3\cdot\frac{1}{2^2}+\cdots\right)^3=8$$

$$S=\frac{8}{3}$$

3. $a+b+c=-a$ $\qquad ab+bc+ca=b$

$\Rightarrow b+c=-2a$ $\qquad$ .....(1)

$ab+c(a+b)=b$ $\qquad$ .....(2)

$c=-2a-b$

$abc=-c$

$c=0$ or $ab=-1$

(i) If $c=0$, $\qquad$ from Equation (1), $b=-2a$

from equation (2), $ab=b$

$$\left.\begin{array}{l}\therefore c=0,b=0,a=0\\ c=0,a=1,b=-2\end{array}\right\}$$

(ii) If $ab=-1\Rightarrow b=-\frac{1}{a}$ & as $bc+a(b+c)=b$

$$c=-2a+\frac{1}{a},\ bc-2a^2=b$$

$$\Rightarrow\left(-\frac{1}{a}\right)\left(-2a+\frac{1}{a}\right)-2a^2=-\frac{1}{a}$$

$$\Rightarrow 2a^4-2a^2-a+1=0$$

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$\Rightarrow (a-1)\left(2a^3+2a^2-1\right)=0$

$a=1$ or $2a^3+2a^2-1=0$

$a=1$ already counted.

$2a^3+2a^2-1=0$ has only one root say $\alpha\in\left(\frac{-2}{3},0\right)$

$\therefore$ only 3 such polynomials

4. $(1-x)^{-2}=1+2x+3x^2+4x^3+5x^4\cdots+(n+1)x^n+\cdots$

$\left((1-x)^2\right)^{-1}=\left(1-2x+x^2\right)^{-1}$

$=\left(1-\left(2x-x^2\right)\right)^{-1}$

$=1+\left(2x-x^2\right)+\left(2x-x^2\right)^2+\cdots+\left(2x-x^2\right)^{n-1}+\left(2x-x^2\right)^n+\cdots$

compare coefficient of x^n in both expansion.

${}^nc_0 2^n-{}^{n-1}c_1 2^{n-2}+{}^{n-2}c_2 2^{n-4}-{}^{n-3}c_3 2^{n-6}\cdots=n+1$

put $n=20$

5. $\sin x=t$

$mt^2+(m-1)t+(m-2)=0$

here (t) is real

$D\geq 0$

$(m-1)^2-4m(m-2)\geq 0$

$m^2+1-2m-4m^2+8m\geq 0$

$-3m^2+6m+1\geq 0$

$3m^2-6m-1\leq 0$

$1-\frac{2}{\sqrt{3}}\leq m\leq 1+\frac{2}{\sqrt{3}}$

Also, roots must lie in [–1, 1]

$m=\frac{t+2}{t^2+t+1}$, min value of m is 1 for solution

$\Rightarrow\ 1\leq m\leq 1+\frac{2}{\sqrt{3}}$

6. Let $a_1=5-d, a_2=5, a_3=5+d, b_1=\frac{3}{q}, b_2=3, b_3=3q$, then the given conditions indicate that

$5-d+\frac{3}{q}$ and $5+d+3q$ are all positive integers and $\left(5-d+\frac{3}{q}\right)(5+d+3q)=64$. It is easy to check that for getting maximum positive d, there are only four possibilities for $\left(5-d+\frac{3}{q},5+d+3q\right)$: (1, 64), (2, 32), (4, 16) and (8, 8).

(i) By solving the system of equations corresponding to (1, 64), it follows that

$3q+\frac{3}{q}=55\Rightarrow q_{1,2}=\frac{55\pm 7\sqrt{61}}{6}, d_{1,2}=4+\frac{3}{q_{1,2}}$

$\Rightarrow d_{max}=4+\frac{3.6}{55-7\sqrt{61}}=4+\frac{55+7\sqrt{61}}{2}$

(ii) By solving the system of equations corresponding to (2, 32), it follows that

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$$q+\frac{1}{q}=8\Rightarrow q_{1,2}=\frac{8\pm\sqrt{60}}{2}=4\pm\sqrt{15}, d_{1,2}=3+\frac{3}{q_{1,2}}$$

$$\Rightarrow d_{max}=3+\frac{3}{4-\sqrt{15}}=15+3\sqrt{15}$$

(iii) By solving the system of equation corresponding to (4, 16), it follows that

$$3q+\frac{3}{q}=10\Rightarrow q=3 \text{ or } \frac{1}{3}, d_{1,2}=1+\frac{3}{q_{1,2}}=2$$

or $10\Rightarrow d_{max}=10$

(iv) By solving the system of equations corresponding to (8, 8), it follows that $q+\frac{1}{q}=2\Rightarrow q=1, d=0$.

In summary, $d_{max}=4+\frac{55+7\sqrt{61}}{2}$, hence the maximum possible value of a_3 is $9+\frac{55+7\sqrt{61}}{2}$

7. $$\frac{(1+x)^{20}+(1-x)^{20}}{2}={}^{20}c_0+{}^{20}c_2x^2+{}^{20}C_4x^4+.....+{}^{20}c_{20}x^{20}$$

$$={}^{20}C_0x^{20}+{}^{20}C_2x^{18}+{}^{20}C_4x^{16}+.....{}^{20}C_{20}$$

So, $\sum_{r=0}^{9}{}^{20}C_{2r}\,{}^{20}C_{2r+2}=\text{coeff of } x^{22} \text{ in } \frac{\left[(1+x)^{20}+(1-x)^{20}\right]^2}{4}=\frac{10}{11}\left({}^{39}c_{21}-{}^{19}c_{10}\right)$

$\Rightarrow a=10\Rightarrow xy=40$ has total order pair (x, y) solution

$xy=40$ has ordered pair as '8' $x,y\in I^+$

$xy=40$ has ordered pair as 16, $x,y\in I$

8. $a_1,a_2,a_3,....a_n$ are in H.P

$\frac{1}{a_1}\frac{1}{a_2}\frac{1}{a_3}.....\frac{1}{a_n}...$ are in A.P with c.d d

$$S=\sum_{i=1}^{n}\frac{n}{\frac{1}{a_{i+1}^{2/3}}+\frac{1}{a_i^{2/3}}+\frac{1}{(a_ia_{i+1})^{1/3}}}$$

$$A=\frac{1}{a_{i+1}^{1/3}}\quad B=\frac{1}{a_i^{1/3}}$$

$$S=\sum_{i=1}^{n}\frac{n}{A^2+B^2+AB}$$

$$S=\sum_{i=1}^{n}\frac{\eta(A-B)}{A^3-B^3}=\frac{n}{d}\sum_{i=1}^{n}(A-B)$$

$$=\frac{n}{d}\sum_{i=1}^{n}\left(\frac{1}{a_{i+1}^{1/3}}-\frac{1}{a_i^{1/3}}\right)$$

$$=\frac{n}{d}\left[\frac{1}{a_{n+1}^{1/3}}-\frac{1}{a_1^{1/3}}\right]=\frac{n}{d}\left[\frac{1}{2}-1\right]$$

$S=-\frac{n}{2d}$ also $\frac{1}{8}=\frac{1}{a_1}+nd.$

$=\frac{8n^2}{14d}=\frac{4n^2}{7}$ $nd=-\frac{7}{8}$

$a=4, b=7$ $d=-\frac{7}{8n}$

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9. Given $3\left(a^2+b^2+c^2+1\right)=2(a+b+c+ab+bc+ca)$

$\Rightarrow (a-b)^2+(b-c)^2+(c-a)^2+(a-1)^2+(b-1)^2+(c-1)^2=0.$

$a=b=c,\ a=b=c=1$

10. Given $x^4-x^3-x^2-1=0.$ roots are a, b, c & d

$g(x)=x^6-x^5-x^4-x^2+x^4-x^3-x$

$=x^2\left(x^4-x^3-x^2-1\right)+x^4-x^3-x \ = x^4-x^3-x^2-1+x^2-x+1 \ = x^2-x+1$

$g(a)+g(b)+g(c)+g(d)$

$=\Sigma a^2-\Sigma a+4 \ =\left((\Sigma a)^2-2\Sigma ab\right)-\Sigma a+4$

$=1+2(-1)-1+4=6.$

11. $\sqrt{10+\sqrt{\gamma}}+\sqrt{10-\sqrt{\gamma}})^2=20+2\sqrt{100-\gamma} \ =2(10+\sqrt{100-r})$

$\because \sum(\sqrt{10+\sqrt{r}}+\sqrt{10-\sqrt{r}})=\sqrt{2}\sum(\sqrt{10+\sqrt{100-r}})$

$\sum \sqrt{10+\sqrt{r}}+\sum \sqrt{10-\sqrt{r}}=\sqrt{2}\sum \sqrt{10+\sqrt{r}}$

as $\sum_{r=1}^{99.}\sqrt{10+\sqrt{100-r}}=\sum_{r=1}^{99}\sqrt{10+\sqrt{r}}$

$\therefore \sum\sqrt{10+\sqrt{r}}+\sum\sqrt{10-\sqrt{r}}=\sqrt{2}\sum\sqrt{10+\sqrt{r}}$

$\Rightarrow \left(\sum\sqrt{10+\sqrt{r}}\right)(\sqrt{2}-1)=\sum\sqrt{10-\sqrt{r}}$

$\Rightarrow \dfrac{\sum\sqrt{10-\sqrt{r}}}{\sum\sqrt{10+\sqrt{r}}}=\sqrt{2}-1$

12. put $x_1=\cot\theta$

$\cot\theta-\tan\theta=2\cot 2\theta \Rightarrow \ x_2=\cos 2\theta$

$x_3=\cot 4\theta$

$x_4=\cot 8\theta$

$x_1=\cot 16\theta.$

$\cot\theta=\cot 16\theta \Rightarrow 16\theta=n\pi+\theta$

$15\theta=n\pi$

$\theta=\dfrac{n\pi}{15} \qquad \theta=\dfrac{\pi}{15},\dfrac{2\pi}{15}\cdots\dfrac{14\pi}{15}$

14 set of solution.

13. $5a^2+5ab=60$

$24ab+36b^2=-60$

Add

$5a^2+29ab+36b^2=0$

$(a+4b)(5a+9b)=0$

$a+4b=0,\ 5a+9b\neq 0$

14. & 15.

$\therefore \ A=\sum_{r=1}^{30}\dfrac{{}^{30+r}C_r(2r-1)}{{}^{30}C_r(30+r)}=\sum_{r=1}^{30}\dfrac{{}^{30+r}C_r}{{}^{30}C_r}\left(1-\dfrac{30-r+1}{30+r}\right)$

$=\sum_{r=1}^{30}\left[\dfrac{{}^{30+r}C_r}{{}^{30}C_r}-\dfrac{{}^{30+r}C_r}{{}^{30}C_r}\cdot\dfrac{30-r+1}{30+r}\right]$

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$$=\sum_{r=1}^{30}\left[\frac{^{30+r}C_r}{^{30}C_r}-\frac{(30+r)}{r}\cdot\frac{^{29+r}C_{r-1}}{^{30}C_r}\cdot\frac{31-r}{30+r}\right]$$

$$=\sum_{r=1}^{30}\left[\frac{^{30+r}C_r}{^{30}C_r}-\frac{^{29+r}C_{r-1}}{^{30}C_{r-1}}\right]$$

$\because \quad \frac{^nC_r}{^nC_{r-1}}=\frac{n-r+1}{r}$

For n = 30,

$\left(\frac{31-r}{r}.{}^{30}C_r={}^{30}C_{r-1}\right)$

$A=\frac{^{60}C_{30}}{^{30}C_{30}}-\frac{^{29+1}C_0}{^{30}C_0}={}^{60}C_{30}-1$, $B={}^{60}C_{30}$, $C={}^{60}C_{30}$

16. & 17.

$P(x)=x^5-9x^4+px^3-27x^2+qx+r$

$\because P(x)$ is divisible by x^2

$\because \quad q=r=0$

$P(x)=x^5-9x^4+px^3-27x^2$

$\frac{P(x)}{x^2}=x^3-9x^2+px-27=0$

$\alpha,\beta,\gamma\in R^+$

$\alpha+\beta+\gamma=9,\alpha\beta\gamma=27$

A.M. of $\alpha,\beta,\gamma=\frac{\alpha+\beta+\gamma}{3}=3$

G.M. of $\alpha,\beta,\gamma=(\alpha\beta\gamma)^{1/3}=(27)^{1/3}=3$

$\because$ A.M. = G.M. $\Rightarrow \alpha=\beta=\gamma$

$\therefore 3\alpha=9\Rightarrow\alpha=3=\beta=\gamma$

$p=\alpha\beta+\beta\gamma+\gamma\alpha=27$

For Q.16. $p+q+r=27+0+0=27$

For Q.17. $\alpha-1=2,\beta+3=6,\gamma+7=10$

$\therefore S_n=2+6+10+\ldots+$ to n terms $=2(1+3+5+\ldots$ to n terms$)=2n^2$

Now, $\sum_{n=2}^{\infty}\frac{1}{\sqrt{S_n\cdot S_{n-1}}}=\sum_{n=2}^{\infty}\frac{1}{\sqrt{2n^2\cdot 2(n-1)^2}}=\frac{1}{2}\cdot\sum_{n=2}^{\infty}\frac{1}{n(n-1)}$

$$=\frac{1}{2}\sum_{n=2}^{\infty}\left(\frac{1}{n-1}-\frac{1}{n}\right)$$

$$=\frac{1}{2}\left(\frac{1}{1}-\frac{1}{2}+\frac{1}{2}-\frac{1}{3}+\ldots\infty\right)=\frac{1}{2}$$

PHYSICS

18. Dimensions of kinetic energy per unit mass = L^2T^{-2}

$$\left[\frac{\alpha\sigma}{\rho}\right]=\frac{ML^{-1}T^{-2}}{ML^{-3}}=L^2T^{-2}$$

19. Consider the disc to be made up of large number of elementary rings. Consider one such ring of radius x & thickness dx, accelerating torque on the elementary ring

$$d\tau_1=dqEx=\sigma(2\pi xdx)\left(\frac{1}{2}\beta x\right)x$$

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$\therefore$ Net accelerating torque

$$\tau_1 = \pi\sigma\beta\int_o^R x^3dx = \frac{\pi\sigma\beta R^4}{4}$$

Similarly, retarding torque

$$d\tau_2 = \eta(2\pi x dx)\left(\frac{\omega x}{t}\right)x$$

$$d\tau_2 = \frac{2\pi\eta\omega}{t}x^3dx \quad \Rightarrow \quad \tau_2 = \frac{2\pi\eta\omega}{t}\int_o^R x^3dx = \frac{\pi\eta\omega R^4}{2t}$$

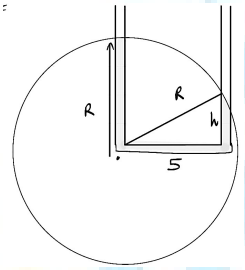
For constant angular velocity

$$\tau_1 = \tau_2$$

$$\frac{\pi\sigma\beta R^4}{4} = \frac{\pi\eta\omega R^4}{2t}$$

$$\Rightarrow \omega = \frac{\sigma\beta t}{2\eta}$$

20.



Potential energy due to liquid will be mainly due to point mass. So liquid will be confined within sphere around it.

$$R + 5 + \sqrt{R^2 - 25} = 25$$

$$-R + 20 = \sqrt{R^2 - 25}$$

$$40R = 425$$

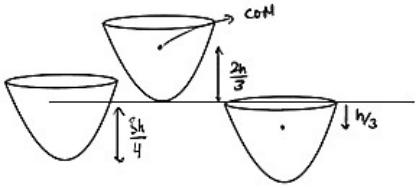
Then $R^2 = h^2 + 25$ will give us $h = \frac{75}{8}$ cm.

21. Pressure $= \frac{2I}{C} = \frac{2}{3\times10^8} \times \frac{\sigma \times 4\pi R^2 \times T^4}{4\pi r^2}$

$$= \frac{2}{3\times10^8} \times \frac{\frac{17}{3}\times10^{-8}\times\frac{1}{4}\times10^{24}}{10^6} = 94.4\ N/m^2$$

22. Conceptual

23.



$$\sigma vg \times h = \rho vg \times \frac{h}{3}$$

$$\frac{\rho}{\sigma} = 3$$

$y = 2x^2 \Rightarrow dy = 4xdx$

At equilibrium, $mg = F_B$

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$$\sigma\times\left(\int_{y=0}^{h}\pi x^2\times dy\right)\times g=\rho g\int_0^{h_1}\pi x^2 dy$$

$$h^2=3h_1^2\Rightarrow h_1=\frac{h}{\sqrt{3}}$$

$$\text{Loss}=mg\times\frac{h}{\sqrt{3}}-mg\times\frac{h}{3\sqrt{3}}=\frac{\sigma\pi gh^3}{6\sqrt{3}}$$

$$ma=\rho\pi\frac{h}{2\sqrt{3}}\times yg$$

$$\frac{\rho\pi h^2}{12}\times a=\frac{\rho\pi hg}{2\sqrt{3}}y$$

24. $P=\sigma\times4\pi R^2\left(T_1^4-T_s^4\right)$

For large sphere

$$P=\sigma\times4\pi\left(R'\right)^2\left(T_1^4-T_2^4\right)$$

Also for lamination

$$P=\sigma\times4\pi\left(R'\right)^2\left(T_2^4-T_s^4\right)$$

$$\Rightarrow T_2^4=\frac{T_1^4+T_s^4}{2}\Rightarrow T_2=\left(\frac{T_1^4+T_s^4}{2}\right)^{1/4}$$

$$R^2\left(T_1^4-T_s^4\right)=\left(R'\right)^2\left[\frac{T_1^4+T_s^4}{2}-T_s^4\right]$$

$$R'=\sqrt{2}R$$

25. $mg\left(h+\frac{H}{2}+\frac{H}{2}\right)=[v\rho g]\frac{H}{2}+[v2\rho g]\frac{H}{2}$

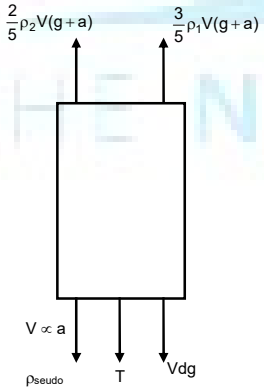
$$h=\frac{7H}{2}$$

26. $h_0=\frac{2S\cos\theta}{r_0\rho g}=\frac{2\times7.5\times10^{-2}}{5\times10^{-4}\times10^3\times10}=\frac{15\times10^{-2}}{5}=3\times10^{-2}\text{m}=30\text{ mm}$

$$h_0r_0=hr$$

$$\Rightarrow r=\frac{h_0}{h}\quad r_0=\left(\frac{30}{25}\right)(0.5)=0.6\text{ mm}$$

27.



We analysed this problem
From the reference frame of
Elevator. Total buoyant force
On the block,

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$$F_B = \left(\frac{2}{5}V\rho_2 + \frac{3}{5}V\rho_1\right)(g+a)$$

For equilibrium,

$$F_B = T + Vd(g+a)$$

or $T = F_B - Vd(g+a)$

$$= (g+a)V\left[\frac{2}{5}\rho_2 + \frac{3}{5}\rho_1 - d\right]$$

$$T = \left(10 + \frac{10}{2}\right)\times 10^{-3}\left[\frac{2}{5}\times 1500 + \frac{3}{5}\times 1000 - 800\right] = 6\ N$$

28. Applying Bernoulli's equation between section A and B

$$P_A + \frac{1}{2}\rho v_A^2 + \rho g h_A = P_B + \frac{1}{2}\rho v_B^2 + \rho g h_B$$

$$P_A + \frac{1}{2}\rho v^2 + \rho g h = P_B + \frac{1}{2}\rho(2v)^2 + 0$$

$$P_B - P_A = \rho g h - \frac{3}{2}\rho v^2 \text{ and } P_B - P_A = \rho g\frac{h}{2}$$

$$\rho g\frac{h}{2} = \rho g h - \frac{3}{2}\rho v2 \Rightarrow v = \sqrt{\frac{gh}{3}}.$$

29. $A_1 + A_2$ and $n_1 + n_2$ are constant

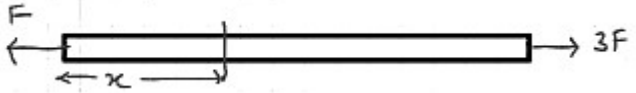
$$\frac{n_1}{n_2} = \frac{r_1^2}{r_2^2} = \frac{4}{1} \Rightarrow r_2^2 = \frac{2R^2}{5}$$

$$r_2 = \sqrt{\frac{2}{5}}R$$

$$r_1 = \sqrt{\frac{8}{5}}R$$

30. $F_{net} = 2F$

$$a = \frac{2F}{m}$$



$$T - F = \frac{mx}{L}\left(\frac{2F}{m}\right) \Rightarrow T = F\left(1 + \frac{2x}{L}\right)$$

Strain energy

$$= \int \frac{T^2}{2AY}, dx = \frac{F^2}{2AY}\int_0^L \left(1 + \frac{2x}{L}\right)^2 dx$$

$$= \frac{F^2}{2AY}\left[L + 2L + \frac{4L}{3}\right]$$

$$U = \frac{F^2}{2AY}\cdot\frac{13L}{3} \Rightarrow \frac{\alpha}{12} = \frac{13}{6}$$

$\alpha = 26$

31. $10^{-3}\times 10 = 2\times 25\times 10^{-3}\times l$

$l = 20$ cm

32. $(m_0 + m)g\times 0.2 = \frac{m_0 g + 3m_0 g}{2}\times 0.2$

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$m_0 + m = 2m_0 \Rightarrow m = m_0$

33. & 34.

Assuming $\lambda = a + bx$

$m = \int \lambda A dx$

$\Rightarrow \int_0^l (a + bx)dx = \frac{m}{A}$

$\Rightarrow al + \frac{bl^2}{2} = \frac{m}{A}$

$\Rightarrow a + \frac{3b}{2} = 900$

Rod is in equilibrium in water

$\int_0^l (a + bx)Adxg \times x = \rho_w Alg \times \frac{l}{2}$

$\frac{al^2}{2} + \frac{bl^3}{3} = \frac{\rho_w l^2}{2} \Rightarrow a + 2b = 1000$

From 1 and 2

a = 600 and b = 200

$I\alpha = \tau_{net}$

$\alpha \times \int_0^1 (a + bx)Ax^2, dx = 1200 \times Alg \times \frac{l}{2} - 1000Alg \times \frac{l}{2}$

$\alpha \times \left[\frac{al^3}{3} + \frac{bl^4}{4}\right] = 200g \times \frac{l^2}{2}$

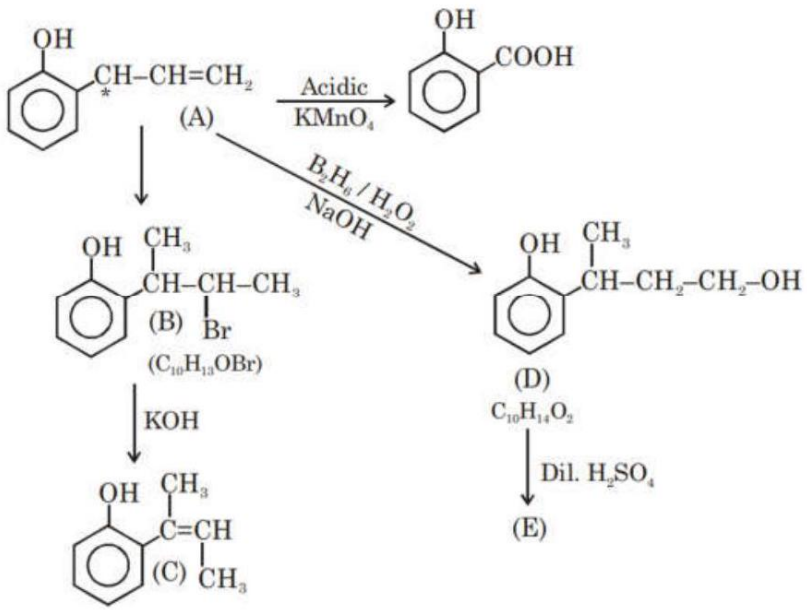
$\alpha = \frac{100g}{\frac{al}{3} + \frac{bl^2}{4}} = \frac{100 \times 10}{200 \times 3 + \frac{200 \times 9}{4}} = \frac{100}{60 + 45} = \frac{100}{105} = \frac{20}{21}$ rad/s^2

CHEMISTRY

35. BrF_5 is square pyramidal $(AX_5E) \rightarrow$ maximum coplanar atoms = 4, number of such planes = 3.

SF_6 (octahedral) and CCl_4 (tetrahedral) have all equilateral triangular faces.

36.



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37. $A(H_2O)_8(s) \rightleftharpoons A(g) + 8H_2O(g); Kp_1$

$(x + y)8x$

$A(H_2O)_8(s) \rightleftharpoons \underset{(x+y)}{A(g)} + 8H_2O(s); Kp_2$

$x + y = 0.2$

$8x = 0.001$

$Kp_1 = (x+y)(8x)^8 = 0.2 \times (10^{-3})^8 = 2 \times 10^{-25}$

$Kp_2 = (x + y) = 0.2$

$H_2O(s) \rightleftharpoons H_2O(g) : \left(\frac{Kp_1}{Kp_2}\right)^{1/8}$

$V.P. = \left(\frac{2 \times 10^{-25}}{2 \times 10^{-1}}\right)^{1/8} = 10^{-3}$ bar

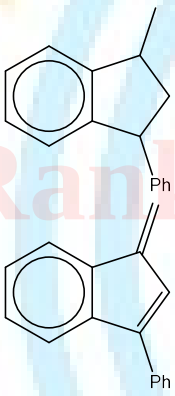
38. $CH_4 + O_2 \xrightarrow[\Delta]{Cu, high\ P} CH_3OH$

$CH_4 + O_2 \xrightarrow[\Delta]{Mo_2O_3} HCHO + H_2O$

$CH_3 - CH_3 + O_2 \xrightarrow[\Delta]{(CH_3COO)_2Mn} CH_3{-}COOH + H_2O$

$CH_4 + \underset{excess}{O_2} \xrightarrow{\Delta} CO_2 + H_2O$

39. A is $Ph - CH = CH_2$

B is 

C is

E is $Ph - C \equiv CH$

40. Conceptual

41. All are correct.

42. a, d, e, f, g & h

43. a, b, c & d

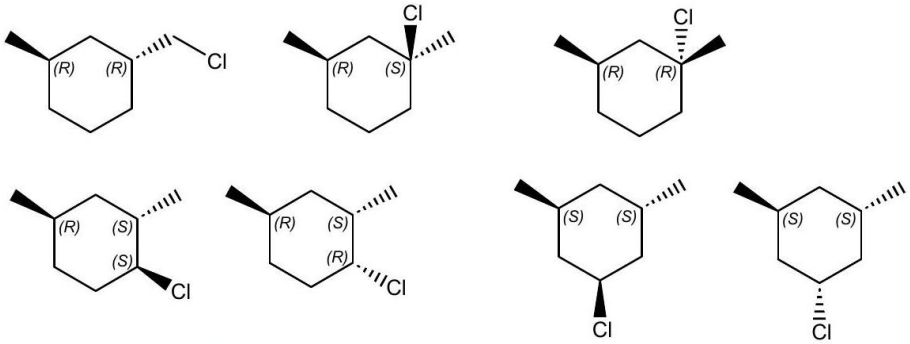
44. a, b, g, h & i

45. all are correct.

46. 2, 4, 5 & 9

47. I = 1, II = 6, III = 3, IV = 8

48. & 49.



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50. & 51.

P_2 (liquid) + Q_2 (gas) ⇌ 2 PQ (gas)

Given:

Kp = 2.5

Vapour pressure of P_2 (liquid) = 0.3 atm

Initially $Q_2 = 1$ mole

At equilibrium, pressure of $PQ = 0.5$ atm

First Question: Find equilibrium pressure of Q_2

For this reaction,

$Kp = (\text{pressure of } PQ)^2 / (\text{pressure of } Q_2)$

So,

$2.5 = (0.5)^2 / P(Q_2)$

2.5=0.25/P(Q2)

Therefore,

$P(Q_2) = 0.25 / 2.5 = 0.1$ atm

So, equilibrium partial pressure of Q_2 = 0.1 atm.

Second Question: Minimum moles of P_2 required

Let x moles of Q_2 react.

From stoichiometry:

1 mole Q 2 reacts → 2 moles PQ form.

So at equilibrium:

Q_2 left = 1 – x

PQ formed = 2x

Total moles at equilibrium = (1 – x) + 2x = 1 + x

Now use pressure ratios.

Total pressure = 0.5(PQ) + 0.1(Q2) = 0.6 atm

Mole fraction of Q2 = pressure of Q_2 / total pressure

= 0.1/0.6

= 1/6

But mole fraction of Q_2 is also:

(1 – x)/(1 + x)

So, (1 – x)/(1 + x) = 1/6

Solving this gives:

$x = 5 / 7 \approx 0.714$

That means 0.714 moles of Q_2 reacted.

Since 1 mole of P_2 reacts with 1 mole of Q_2, the amount of P_2 required is also 0.714 moles. So the minimum moles of P_2 needed initially = 0.714 moles (approximately 5/7 moles).

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